

TrendsTalk

Scientific mobility in microbiology – 6

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Can you tell us about your research interests and scientific background?

My research group at The Sainsbury Laboratory (TSL) investigates how individual plant and microbial cells interact. To uncover the molecular and cellular dynamics underlying these interactions, we develop and apply novel genomics and imaging technologies. For example, using single-cell and spatial omics approaches, we recently discovered that plants deploy spatially organized immune cell states in response to pathogen attack [1].

My academic journey has taken me across several countries. I completed my undergraduate studies at the University of Tokyo in Japan, followed by a one-year master's internship at CEA Cadarache in France. I then pursued my PhD at the Max Planck Institute for Plant Breeding Research in Germany and continued as a postdoctoral researcher at the Salk Institute in the USA. Throughout these experiences, I have remained broadly interested in plant and microbial biology, with a particular passion for advancing new technologies to study their interactions.

What have been some of the biggest mobility-related hurdles and challenges that you have faced during your academic career?

I have been fortunate not to face major barriers when moving between countries, but even smooth transitions come with a series of small, often stressful challenges, such as visa applications, healthcare access, housing, and financial arrangements. These tasks are especially demanding when they coincide with starting a new role, as was the case for me when beginning my PhD, postdoc, and principal investigator (PI) positions.

One particular challenge was relocating with our two cats – from Japan to the USA, and then from the USA to the UK. Each move involved navigating entirely different animal import regulations, timelines, and paperwork, which added logistical complexity. For example, moving from Japan to the USA was relatively straightforward in terms of regulation – there was no quarantine requirement, and our cats could fly in-cabin after we obtained a basic health certificate from a government-approved veterinarian. However, this move happened during the coronavirus disease 2019 (COVID-19) pandemic, which made scheduling flights and securing appointments more difficult. By contrast, the move from the USA to the UK was more complex: the UK requires microchipping, a rabies vaccination given at least 21 days before travel, and a US Department of Agriculture (USDA)-endorsed health certificate issued within 10 days of entry. In our case, the cats had to fly in the cargo hold, which added logistical challenges and stress. For more information, see UK pet travel rulesⁱ and USDA export requirementsⁱⁱ.

Another challenge was the visa process. When we moved to the UK, in 2024, we applied for a 5-year visa for two people, which required a substantial upfront payment, including the healthcare surcharge, which had risen significantly in the previous 10 years. Without institutional support, these costs could have made the move difficult.

